Lattice Signature with Efficient Protocols

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1 Introduction

Digital signatures are fundamental cryptographic primitives, allowing a user to authenticate a message and guarantee its integrity. For decades, most practical signature schemes have relied on classical hardness assumptions such as integer factorization or discrete logarithms. However, these assumptions are known to be broken by quantum algorithms, which motivates the search for alternative constructions secure in the post-quantum setting. Lattice-based cryptography has emerged as one of the most promising frameworks for post-quantum security, as its underlying problems are believed to remain hard even for quantum computers. Beyond security, modern applications often require signatures to interact efficiently with zero-knowledge proof systems, for instance in privacy-preserving protocols. Designing lattice-based signatures that remain compatible with such proof systems is therefore crucial. A first attempt in this direction was the construction of Libert, Ling, Mouhartem et al. ([2]), which proposes a lattice-based signature scheme with efficient protocols. Although conceptually appealing, their construction suffers from extremely large proof sizes (several hundreds of megabytes) making it impractical. The paper studied in this report ([1]) revisits this problem and proposes a new lattice-based signature scheme specifically designed to interface smoothly with recent zero-knowledge argument systems, while reducing the size of the produced proofs. In the following, we analyse the main ideas behind this construction, highlight its improvements over previous work, and examine some of its technical components in more detail.

2 General Paper Analysis

The goal of the paper by Jeudy, Roux-Langlois and Sanders ([1]) is the design of lattice-based signatures that work well with zero-knowledge proof systems, a property known as "signatures with efficient protocols" (SEP). This is crucial because privacy-preserving authentication systems — like anonymous credentials, group signatures, e-cash and so on — require proving knowledge of a signature on a hidden message and because the systems deployed up to now revealed to be vulnerable to quantum computers.

One of the prior lattice-based SEP constructions is Libert et al. (Asiacrypt 2016) ([2]). In the paper, the authors found a way to make lattice-based signatures work with zero-knowledge proof systems. However, their proofs of knowledge reached hundreds of megabytes (the paper provides a fair and quantitative

analysis of this limitation). All cited works appear in top-tier venues (Crypto, Eurocrypt, Asiacrypt, Journal of Cryptology), and the paper itself was published at Crypto 2023.

The paper studied in this report ([1]) aims to fill in the gap caused by the problem of these previous works. In order to do this, they generate the signature and commitment parameters together. This one change makes a difference: fewer witnesses, smaller SIS bounds, and much better parameters.

The methodology builds on well-established tools — Micciancio-Peikert trapdoors ([3]), Rényi divergence, and the Yang et al. ([4])/ LNP22 zero-knowledge frameworks — and is technically sound. The paper also identifies and corrects an error in the fast mode of Yang et al., which is a valuable independent contribution.

The results are amazing. The proof sizes go down from around 670 MB to below 650 KB in the module instantiation, a reduction of over three orders of magnitude.

The weaknesses are not fundamental in the sense that they arise from the methodology rather than from insurmountable barriers. The main practical weakness is the remaining proof size overhead relative to classical constructions, but it simply reflects the general current state of lattice-based zero-knowledge proofs. It is plausible that further advances in proof systems will reduce these sizes further, and the paper's modular design allows to directly benefit from such future improvements.

The paper has attracted follow-up citations in the lattice cryptography literature, with works using SEP framework from this paper as a part of their considerations, consistent with the authors' stated goal of providing a versatile building block. These works generally affirm the paper's contributions and have not identified fundamental flaws, though some note on further efficiency improvements that could be done improve their constructions as well.

This paper, presented at Crypto 2023, was the first one to provide a practical way to create anonymous credentials that can withstand quantum computers. It is likely to remain influential as long as lattice-based cryptography remains the dominant post-quantum paradigm.

3 Paper Contributions

3.1 Large Sizes Problem

As mentioned in the introduction, the signature scheme from Libert et al. [2] suffers from large sized proofs (30 to 40 times larger). In this section, we study why this previous scheme yields such enormous sizes. In [2], the signature scheme relies on a commitment $\mathbf{c} = \mathbf{D_0 r} + \mathbf{D_1 m}$. To prove knowledge of a valid signature on a hidden message, the prover must demonstrate knowledge of the opening $(\mathbf{r}, \mathbf{m})$ such that $\mathbf{c}$ is well formed. More precisely, the verification algorithm operates on the binary decomposition of the commitment vector. In our case, $\mathbf{c} \in \mathbb{Z}_q^n$ so the protocol requires decomposing each coefficient c_i into $l = \lceil \log_2 q \rceil$ bits.
This design choice induces 2 main inefficiencies :

1. Witness expansion : The witness size for the ZKAoK scales linearly with

the number of bits to be proven. Instead of proving knowledge of n elements in Z_q^n, the prover must prove knowledge of $n \cdot \log_2 q$ binary variables.

2. Constraint proliferation : To ensure the decomposition is valid, the ZK protocol must enforce the binary constraint for every bit. In frameworks based on quadratic constraints, this requires adding $\mathcal{O}(n \log q)$ quadratic equations to the proof system. Furthermore, the security reduction in [2] relies on embedding a trapdoor within the decomposition matrix. This introduces a norm growth factor $\propto \sqrt{n \log q}$ in the extracted solution, forcing the scheme to adopt overly conservative parameters to maintain security margins.

3.2 The Solution

To mitigate this, we optimize the interactions between the commitment scheme implicitly used by privacy-enhancing protocols and the signature scheme itself. Instead of generating the public parameters of these 2 components independently from one another, the authors propose a unified generation, using a common matrix $\mathbf{A}$. This indeed reduces the size of the proofs since the size of the public key is smaller and it allows the commitment randomness $\mathbf{r}$ to be algebraically absorbed into the signature vector $\mathbf{v}$, eliminating the need for a separate proof of opening.

4 Signature Scheme

The signature scheme proposed by the paper is based on the one studied in [2], along with the improvements discussed in the previous section.

The proposed construction is a lattice-based signature scheme with efficient protocols, designed to minimize the witness size required for subsequent zero-knowledge proofs. While it inherits the high-level goal of [2] (signing committed messages), its internal mechanics differ fundamentally to achieve the efficiency gains outlined in Section 2.2. The scheme relies on the hardness of the Short Integer Solution (SIS) problem over lattices and utilizes the trapdoor sampling framework of Micciancio and Peikert [3].

4.1 Key Generation and Parameters

The scheme operates over a modulus q and dimensions n, m. The key generation process (Algorithm 3.2 in [1]) produces a public key $pk = (\mathbf{A}, \mathbf{B}, \mathbf{u})$ and a secret key $sk = \mathbf{R}$.

- $\mathbf{A} \in \mathbb{Z}_q^{n \times m_1}$ is chosen uniformly at random. Crucially, in the context of signing hidden messages, $\mathbf{A}$ is derived from a public seed via a hash function (modeled as a Random Oracle) to ensure it is statistically random and independent of the trapdoor.
- $\mathbf{R} \in \{-1, 0, 1\}^{m_1 \times m_2}$ is a short random matrix serving as the *trapdoor*.
- $\mathbf{B} = \mathbf{AR} \pmod q$ is computed such that the relation between $\mathbf{A}$ and $\mathbf{B}$ is hidden by the shortness of $\mathbf{R}$.

- $\mathbf{u} \in \mathbb{Z}_q^n$ is a random syndrome defining the specific lattice coset for signatures.

The public parameters also include a gadget matrix $\mathbf{G}$ and a commitment key $\mathbf{D}$, which are used during the signing and verification phases.

4.2 Signing Committed Messages

The core innovation lies in the signing algorithm (Algorithm 3.3). To sign a binary message $\mathbf{m}$ hidden inside a commitment $\mathbf{c}$, the protocol proceeds as follows:

1. **Commitment:** The user computes a commitment to $\mathbf{m}$ using the public matrix $\mathbf{A}$:
$$\mathbf{c} = \mathbf{A}\mathbf{r} + \mathbf{D}\mathbf{m} \pmod{q} \tag{1}$$
where $\mathbf{r}$ is a Gaussian randomness vector. Note that $\mathbf{A}$ is shared between the signature key and the commitment scheme.

2. **Preimage Sampling:** The signer chooses a tag τ and uses the trapdoor $\mathbf{R}$ to sample a short vector $\mathbf{v}'$ satisfying a modified linear equation:
$$[\mathbf{A} \mid \tau\mathbf{G} - \mathbf{B}]\mathbf{v}' = \mathbf{u} + \mathbf{c} \pmod{q} \tag{2}$$
Substituting $\mathbf{c} = \mathbf{A}\mathbf{r} + \mathbf{D}\mathbf{m}$ and $\mathbf{B} = \mathbf{A}\mathbf{R}$, the equation becomes:
$$\mathbf{A}\mathbf{v}_1' + (\tau\mathbf{G} - \mathbf{A}\mathbf{R})\mathbf{v}_2' = \mathbf{u} + \mathbf{A}\mathbf{r} + \mathbf{D}\mathbf{m} \tag{3}$$

3. **Randomness Merging:** The final signature vector $\mathbf{v}$ is constructed by subtracting the commitment randomness $\mathbf{r}$ from the first part of $\mathbf{v}'$:
$$\mathbf{v} = \mathbf{v}' - \begin{bmatrix} \mathbf{r} \\ \mathbf{0} \end{bmatrix} \tag{4}$$
The resulting signature is the pair $\sigma = (\tau, \mathbf{v})$.

4.3 Verification

Verification (Algorithm 3.4) checks the validity of the signature without knowing $\mathbf{r}$. The verifier computes:

$$[\mathbf{A} \mid \tau\mathbf{G} - \mathbf{B}]\mathbf{v} \stackrel{?}{=} \mathbf{u} + \mathbf{D}\mathbf{m} \pmod{q} \tag{5}$$

By substituting $\mathbf{v} = \mathbf{v}' - [\mathbf{r}^T \mid \mathbf{0}^T]^T$, the term $\mathbf{A}\mathbf{r}$ from the commitment cancels out with the $\mathbf{A}\mathbf{r}$ implicitly contained in $\mathbf{v}'$. This algebraic cancellation is the key mechanism: it allows the verifier to check the signature against the message $\mathbf{m}$ directly, without needing to verify the commitment opening $\mathbf{r}$ separately. Additionally, the verifier checks that $\|\mathbf{v}\|$ is sufficiently small to ensure security against lattice reduction attacks.

4.4 Security Considerations

The reuse of matrix $\mathbf{A}$ for both the signature trapdoor and the commitment introduces a potential vulnerability: if the signer could choose $\mathbf{A}$ maliciously, they might break the hiding property of the commitment. To prevent this, the scheme requires $\mathbf{A}$ to be generated via a hash function (Random Oracle Model) or a trusted setup, ensuring $\mathbf{A}$ is statistically uniform and independent of the trapdoor $\mathbf{R}$. The security of the scheme is proven under the SIS assumption, distinguishing between Type I forgeries (new tag) and Type II forgeries (reused tag), with the latter requiring the use of Rényi divergence to handle the noise distribution introduced by the randomness merging.

5 Technical Analysis

5.1 The `SampleD` Algorithm

The main advantage of this algorithm is its efficiency. Hence, we can easily try to make it work for larger dimensions. The contributors published its implementation on `github.com/truncatedsampler/truncated-sampler`. Unlike previous instantiations that relied heavily on spherical Gaussians, the newer approaches utilize elliptical Gaussian distributions to further compress the signature size and speed up the generation. This optimization is crucial for reducing the required bounds in the underlying Short Integer Solution (SIS) problem and improving the overall efficiency of the pre-image sampling.

5.2 The Signature Scheme

At the core of the proposed construction is a radical redesign of how the signature interacts with the message commitment. In previous state-of-the-art models like [2], the commitment to a hidden message required a binary decomposition, which severely deteriorated the quality of the extracted SIS solution and drastically inflated the Zero-Knowledge Proof (ZKP) size.

To solve this issue, the authors propose a much more compact verification equation:

$$[A|\tau G - B]v = u + Dm \pmod q \tag{6}$$

where A and B are public matrices, G is the gadget matrix, τ is a unique tag, u is a random syndrome, D is the commitment key, and v is the short signature vector.

To sign a hidden message m, the user first commits to it via $c = Ar + Dm$, where r is a Gaussian randomness. The central innovation of this scheme is the absorption of this randomness r directly into the pre-image v' generated by the `SampleD` algorithm. By setting the final signature vector as $v = v' - [r^T|0]^T$, the protocol removes the need to separately prove the knowledge of the commitment opening. This algebraic simplification has a cascading effect, removing thousands of intermediate witnesses in the subsequent zero-knowledge proof.

Furthermore, the protocol utilizes the tag τ to prevent replay attacks and ensure uniqueness. In privacy-preserving applications (such as anonymous credentials or e-voting systems), this stateful tag acts as a serial number that

guarantees a user cannot double-spend or double-vote, while keeping their actual message m totally secret.

Finally, while the standard lattice implementation reduces the proof size to approximately 18 MB, the authors push the efficiency further by instantiating the scheme over structured lattices, specifically module-SIS and module-LWE. Coupled with recent exact ZK proof frameworks (such as the Lyubashevsky-Nguyen-Plançon [**LNP22**] framework), this approach reduces the size of a proof of possession of a message-signature pair to less than 650 KB. This monumental decrease makes the signature scheme highly practical for real-world, privacy-enhancing applications.

5.3 Application to E-Voting Systems

The technical properties of the proposed signature scheme make it particularly advantageous for electronic voting protocols. In a secure e-voting scenario, the fundamental requirements are voter privacy (anonymity), vote unforgeability, and public verifiability of the tally. The stateful tag τ integrated into the verification equation provides a mathematically sound mechanism to prevent double-voting. Specifically, a unique tag can be deterministically derived for each voter per election. When a vote is cast, the tag is revealed, allowing voting authorities to verify that no voter has submitted multiple ballots, all without compromising the secrecy of the committed vote m.

Moreover, the drastic reduction in the size of the zero-knowledge proofs ensures that the protocol remains highly scalable. In a large-scale election, storing and verifying millions of proofs becomes completely feasible, addressing one of the main bottlenecks of previous lattice-based privacy-preserving systems. The combination of post-quantum security guarantees, efficient pre-image sampling, and compact proof sizes establishes this scheme as a robust foundation for next-generation e-voting infrastructures.

6 Work Organization

First, the members of the group studied the paper on their own, in order to have a good understanding of the content. A meeting was then hold in order to exchange any questions and to give a better comprehension of the article to every member of the group. As for the work distribution, 2 groups have been made, one group (3 members) focusing on the theoretical part and one group (2 members) focusing on the practical part. Even though the groups are studying different branches of the project, it is still very important to understand everything that is being done, therefore, meetings are and will be organized to share what the students did with eachother.

7 Conclusion

In conclusion, the paper makes some important contribution to lattice-based cryptography by addressing a challenge that is both technical and practical : how to design a post-quantum signature scheme that remains efficient in privacy-preserving applications. Rather than making a simple optimization, the authors redesign the interaction between the signature and the commitment

mechanism, significantly reducing the size of the zero-knowledge proofs required to prove possession of a valid signature on a hidden message.

What makes the construction particularly interesting is that these efficiency gains do not come at the cost of security. The scheme still relies on standard lattice assumptions, like SIS, while using trapdoor sampling and careful parameter design. The paper shows that efficiency in modern cryptography depends not only on the security of a primitive itself, but also on how well it interacts with other components such as commitments and zero-knowledge proofs. More generally, it shows that lattice-based constructions can be made much more practical for real-world privacy applications.

Until now, we mainly focused on getting a solid grasp on the paper subject, and trying to understand the code written to implement the signature scheme by the authors. We also analysed the paper more generally by looking at its impact in the field, how it is written, the methodology used by the authors, ... In the second half of the project, we would like to use the code from the authors in order to verify the sizes claimed for the signatures, maybe implement our own small version of the signature scheme using a high-level language. As for the theoretical part, our goal is to dive into the proofs to understand deeply their impact and how they are built.

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